

3 The cell membrane determines which molecules can enter or leave a cell.

(a) Which pair of membrane transport mechanisms only involves a **solute** moving down a concentration gradient?

(1)

- ☐ A active transport and diffusion
- ☐ B active transport and osmosis
- ☐ C diffusion and facilitated diffusion
- ☐ D facilitated diffusion and osmosis

(b) Which of the following transport mechanisms involve carrier proteins in the membrane?

- active transport
- diffusion
- exocytosis

(1)

- ☐ A active transport only
- ☐ B active transport and exocytosis
- ☐ C diffusion only
- ☐ D diffusion and exocytosis

(c) Which cell transport mechanism is used to take large particles or bacteria into a cell?

(1)

- ☐ A active transport
- ☐ B endocytosis
- ☐ C exocytosis
- ☐ D facilitated diffusion

(d) Which cell transport mechanism moves small lipid-soluble molecules through the membrane?

(1)

- ☐ **A** active transport
- ☐ **B** diffusion
- ☐ **C** facilitated diffusion
- ☐ **D** osmosis

(e) Channel proteins are involved in facilitated diffusion.

Which row of the table describes the charge on the R groups on the amino acids of a channel protein?

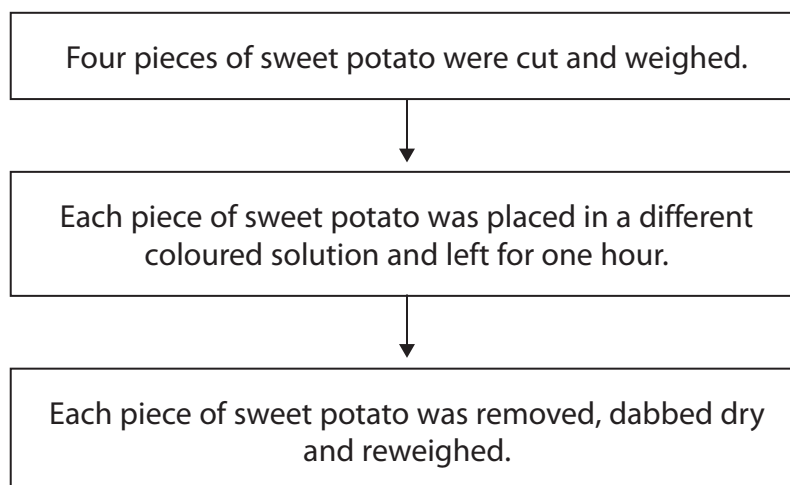
(1)

	Charge on R groups facing inside the channel are mostly	Charge on R groups facing the rest of the membrane are mostly
☐ A	polar	polar
☐ B	polar	non-polar
☐ C	non-polar	polar
☐ D	non-polar	non-polar

(f) A student investigated the effect of four different coloured solutions on the mass of pieces of sweet potato.

Dyes were dissolved in water to produce the coloured solutions.

The diagram shows the method used by the student.

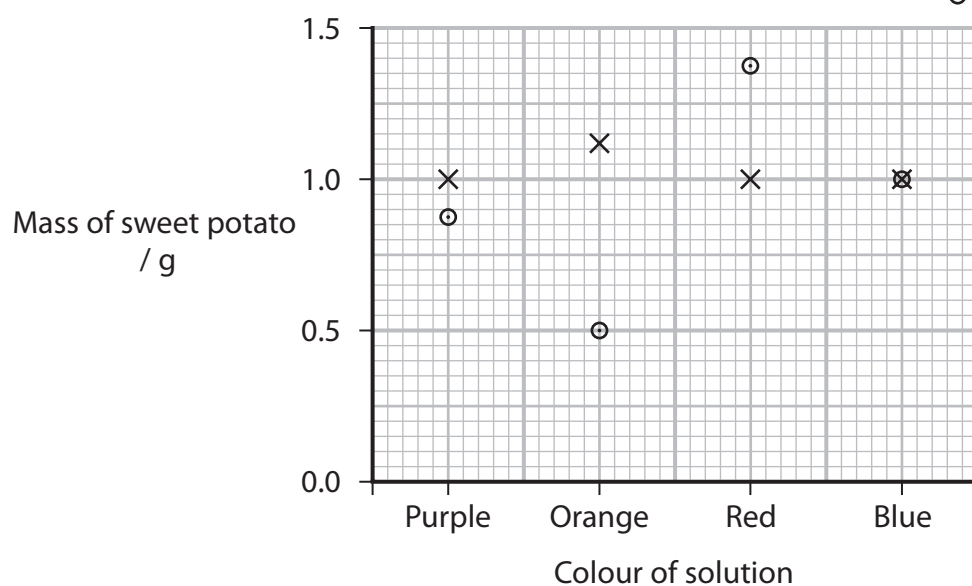


The graph shows the results of this investigation.

Key

× initial mass of sweet potato

○ final mass of sweet potato



Explain the results of this investigation.

(4)

(Total for Question 3 = 9 marks)

